Two-body low-energy scattering

Theory and implementation: every function, every constant, and why

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Abstract

We solve the s-wave radial equation for two particles at low energy and extract the scattering length a, the effective range r_0 and the bound-state energy, for four model potentials and four measured helium-helium potentials. Each numerical constant is derived or measured here rather than asserted, each approximation is given the magnitude of the error it costs, and each result is checked against a closed form or a published value. The record of how the code changed over time is in its version history and is not reproduced here.

Contents

1 What universality means here

At low energy, when the de Broglie wavelength is comparable to or larger than the range of the potential, the detailed shape of the interaction becomes invisible. Only two numbers survive — the **scattering length** a and the **effective range** r_0 — through the effective-range expansion

$$k \cot \delta_0(k) \;=\; -\frac{1}{a} \;+\; \frac{1}{2} r_0 \, k^2 \;+\; O(k^4). \tag{1}$$

This is what universality means: the same two-parameter description applies across physical domains that have nothing else in common. In the words of Ref. [3], *"the universal behavior in the low-energy limit enables us to draw a parallel between cold atomic gases and nuclear systems."*

The two systems used throughout this work make the point concrete:

system	a	r_0	E (measured)	domain
^{4}He dimer	90.4 Å	8.0 Å	−1.62 mK	atomic
deuteron	5.4112 fm	1.7436 fm	−2.224 MeV	nuclear

Both quantities must be converted to common units before they are compared, which is the whole point of the exercise. With $k_B = 8.617 \times 10^{-5}$ eV/K, the deuteron sits at 2.224×10^6 eV and the helium dimer at 1.396×10^{-7} eV: a ratio of 1.59×10^{13}, so **13 orders of magnitude in energy**. In length, 90.4 Å against 5.4112 fm is 1.67×10^6, so **6 orders**. One is held together by van der Waals and the other by the strong interaction, and yet the same two numbers, fed into the same two formulas, reproduce both binding energies.

The two figures are not independent. If $E \sim \hbar^2/(2m_r a^2)$, the energy ratio should be the length ratio squared times the reduced-mass ratio: $(1.67 \times 10^6)^2 \times 3.97 = 1.11 \times 10^{13}$, against 1.59×10^{13} measured. The factor of 1.44 left over is what the effective range adds on top of the zero-range estimate. So the 13 orders in energy *are* the 6 in length, squared, plus the masses — which is the scaling law working, not a coincidence.

Shape independence is the mechanism, not the definition

A separate — and often confused — statement is that *different potentials* tuned to the same (a, r_0) give the same low-energy observables. That is not the definition of universality; it is a consequence of the **shape-independent approximation**, Eq. (55) of [3], which says that $\delta_0(k)$ at low energy does not depend on the shape of the potential. As [3] puts it, the fact that four very different potentials can be tuned to the same a and r_0, *"although remarkable, is directly related to the shape-independent approximation."*

So the roles are:

universality	two parameters describe low-energy physics across domains
shape independence	why the shape of V beyond those two numbers does not matter
tuning	the numerical procedure that puts a given V at a chosen (a, r_0)

Why: Tuning is a *demonstration technique*, not a physical claim. The tuned parameters are of course different for each potential — a Gaussian and a Lennard-Jones that share (a, r_0) have nothing numerically in common. What they share is everything a low-energy experiment can measure.

When two parameters are not enough

The expansion (1) is a series, and dropping r_0 means keeping only its first term. Whether that is legitimate depends on how small kR is. Reference [3] makes exactly this point for the deuteron: the zero-range formula *"only accounts for* $\approx 64\%$ *of the deuteron energy"*, because *"$kR \sim 0.4$ for the deuteron is larger than in the ^{4}He case."*

system	$r_0/\lvert a\rvert$	zero range	finite range
deuteron	0.32	−36%	−0.05%
^{4}He dimer	0.089	−8.6%	+0.47%

Why: The deuteron is the textbook example of a shallow bound state, and it is the one that needs the finite-range correction most. Being universal is not a property a system has or lacks in the abstract; it is a statement about how small the expansion parameter happens to be.

2 Conventions and units

The s-wave radial Schrödinger equation, written for $u(r) = r\,\psi(r)$, is

$$-\frac{\hbar^2}{2m_r}\,u''(r) + V(r)\,u(r) = E\,u(r), \tag{2}$$

with $m_r = m_1 m_2/(m_1+m_2)$ the reduced mass. The code works with rescaled quantities, defined explicitly as

$$V_{\text{code}} = \frac{m_r}{\hbar^2}\,V_{\text{phys}}, \qquad E_{\text{code}} = \frac{m_r}{\hbar^2}\,E_{\text{phys}}, \tag{3}$$

which turns the equation into

$$\boxed{u'' = 2\left(V-E\right)u} \tag{4}$$

the only equation the code ever solves, at $E = 0$ and at $E < 0$ alike.

Why: A common shorthand for this is "$\hbar = m_r = 1$, lengths in fm". It is wrong twice over.

It fixes a unit the solver does not fix. The integrator works on pure numbers. Whatever length unit the potential's parameters were written in is the unit that R, a and r_0 come back in. Call it ℓ — not L, which is already the rescaling factor of Sec. 16.1 and a different thing. Two choices appear in this document and nothing in the solver distinguishes them: ℓ = fm for the four model potentials, whose parameters come from the nuclear tables of [3], and ℓ = Å for the measured helium potentials of Sec. 20, published in kelvin and ångström. The unit rides in with the potential and rides out with the answer.

It suggests V and E are pure numbers, and they are not. Distances are never rescaled, so Eq. (3) gives

$$[V_{\text{code}}] = [E_{\text{code}}] = \ell^{-2}.$$

Everything downstream follows from that single line: the units of the Lennard-Jones coefficients in Sec. 5, $[C_6] = \ell^4$ and $[C_{12}] = \ell^{10}$, and with them the exponents of the scaling rule in Sec. 16.1. Treating V as dimensionless is exactly how the wrong exponent $C_{12} \to C_{12}L^6$ reached an earlier draft.

Back to physical units

Comparing (4) with the dimensional form,

$$\frac{2m_r}{\hbar^2}\big(V_{\text{phys}} - E_{\text{phys}}\big) = 2\big(V_{\text{code}} - E_{\text{code}}\big) \implies E_{\text{phys}} = \frac{\hbar^2}{m_r}\,E_{\text{code}},$$

and since $\hbar^2/2m_r$ is the constant usually tabulated,

$$\boxed{E_{\text{phys}} = 2\left(\frac{\hbar^2}{2m_r}\right)E_{\text{code}}} \tag{5}$$

Why: The factor of 2 is checked in Sec. 16: with it wrong, the inferred constant would not come out at 41.47 MeV·fm^2.

The potential converts by the same rule, and it is worth writing separately because it is what the Lennard-Jones discussion of Sec. 5 turns on:

$$V_{\text{phys}}(r) = \frac{\hbar^2}{m_r}\,V_{\text{code}}(r) = 2\left(\frac{\hbar^2}{2m_r}\right)V_{\text{code}}(r). \tag{6}$$

3 Nondimensionalisation

Section 2 rescales energies but leaves lengths carrying a unit, which is why $[V_{\text{code}}] = \ell^{-2}$ rather than being pure numbers. This section removes the remaining unit.

3.1 The transformation

Start from the dimensional equation,

$$-\frac{\hbar^2}{2m_r}\,u''(r) + V(r)\,u(r) = E\,u(r). \tag{7}$$

Choose any length r_* and set $x = r/r_*$. Since $\mathrm{d}^2/\mathrm{d}r^2 = r_*^{-2}\,\mathrm{d}^2/\mathrm{d}x^2$,

$$\frac{\mathrm{d}^2u}{\mathrm{d}x^2} = \frac{2m_r r_*^2}{\hbar^2}(V - E)u. \tag{8}$$

The prefactor is the reciprocal of an energy, and that energy is the one r_* implies:

$$\boxed{\varepsilon_* \equiv \frac{\hbar^2}{2m_r r_*^2}} \tag{9}$$

Writing $v(x) = V(r_*x)/\varepsilon_*$ and $e = E/\varepsilon_*$,

$$\boxed{u''(x) = \big[v(x) - e\big]\,u(x)} \tag{10}$$

in which every symbol is a pure number.

Why: The factor of two is gone, and that is the point of doing this. Equation (4) carries $u'' = 2(V - E)u$, and the 2 is there because that convention divides energies by $\hbar^2/m_r$ while the length that would have made the equation natural, $\hbar^2/2m_r$, is the one the literature tabulates. Section 2 needs a boxed warning that the factor is easy to lose. In Eq. (10) there is no factor to lose.

3.2 Why r_* is not physics

r_* is a bookkeeping choice, not an input. It has to cancel from every measurable quantity, and that is checkable rather than assumed: solve Eq. (10) with different r_* and the dimensionless outputs must scale so that $a = r_* \, (a/r_*)$ is the same number.

r_* (Å)	what it is	$x_{\max}$	a/r_*	a (Å)
1.000	arbitrary	1700.36	88.439115	88.439115
2.963	r_m of the potential	573.86	29.847828	88.439115
10.00	arbitrary	170.04	8.843911	88.439115
64.60	r_{vdW} of ^{39}K	26.32	1.369027	88.439115

Aziz HFD-B on the ^{4}He dimer, with the cut taken at $|v(x_{\max})| \, x_{\max}^2 = 10^{-10}$. Four choices of r_* spanning a factor of 65, and a agrees to eight figures. r_0 likewise, at 7.276317 Å in all four.

Why: This is what the previous convention could not do. With the cut written as $|V(R)| = 10^{-12}$ — a bare number carrying ℓ^{-2} — the same physical potential gives $R = 273$ Å when its parameters are written in ångström and $R = 5.97$ Å when they are written in fm, and the answers differ. The dimensionless cut $|v|x^2 = \varepsilon$ removes that, because $[v] = 1$ and $[x] = 1$.

3.3 What is fixed and what is chosen

fixed by the physics	chosen by us
ε_*, once r_* is chosen, Eq. (9)	r_*, any length
the shape of $v(x)$	ε, where the tail is cut
a/r_0, and the sign of a	the grid, `POINTS`

The left column is what a measurement can see. Note what is *not* in it: neither a/r_* nor r_0/r_* is physical on its own, because r_* is ours. The two-body problem at low energy has exactly one dimensionless number, r_0/a, together with the sign of a — which is the same quantity Sec. 18.3 argues should be reported and Sec. 11.1 uses to decide when a counts as infinite. It arrives here for the third time, now as a consequence of the transformation rather than as a suggestion.

3.4 Verification against the closed forms

The square well keeps its exact range, $R = 1/\mu$, so taking $r_* = R$ puts the edge at $x_{\max} = 1$ and the well depth becomes the single number $-2v$.

	dimensionless	recovered	closed form
a	$2.69993651 \, r_*$	5.39987301	5.39987301
r_0	$0.84890211 \, r_*$	1.69780422	1.69780422

Deuteron parameters, $r_* = 1/\mu = 2$ fm. Against Eqs. (80) and (92) of [3] the errors are 1.6×10^{-10} and 1.1×10^{-11}, the same accuracy the dimensional code reaches.

3.5 What this section does not settle

Three things are open, and stating them is more useful than implying they are not.

The value of ε is now measured, and it is not what the old one translates to. Tightening the tail cut makes the answer *worse*. The r_0 integrand carries a factor r, so pushing R outward multiplies the accumulated round-off by r. On the deuteron Lennard-Jones at 8001 points:

ε	r_0	under refinement
10^{-6}	1.699102	stable to 10^{-8}
10^{-8}	1.699102	stable to 10^{-6}
10^{-10}	1.699796	drifting, 4×10^{-5}
10^{-12}	1.707	wrong
10^{-14}	-0.156	and $+10.4$, then -30.2, as the grid refines

$\varepsilon = 10^{-6}$ sits in the middle of the stable window. The core cut is converged from 10^4 upward. Neither number could have been guessed from the dimensional values they replace.

The inner start. The outer cut is dimensionless above; the inner one is not. The test that produced these tables began integration at a fixed physical 0.30 Å, which is exactly the kind of hidden dimensional constant this section exists to remove. A dimensionless criterion for $x_{\min}$ is still needed.

Whether r_* should be global or per potential. The results do not depend on it, as the table shows. The *conditioning* might: with $r_* = 64.6$ Å the grid spans $x \in [0.005, 26]$, and with $r_* = 1$ Å it spans $[0.3, 1700]$ — two and a half decades apart in dynamic range. Both agree to eight figures for this potential. Whether that survives a stiffer one has not been tested.

4 The one idea: do not integrate what is already known

Every potential here has a range R. Beyond R we have $V = 0$, and there Eq. (4) has a solution known without computing anything:

$$E = 0: \quad u'' = 0 \qquad \Longrightarrow \qquad u \text{ is a } \textbf{straight line}, \tag{11}$$

$$E < 0: \quad u'' = -2E\,u \qquad \Longrightarrow \qquad u \propto e^{-\kappa r}, \quad \kappa = \sqrt{-2E}. \tag{12}$$

So the code **never integrates past** R. It integrates from the inner boundary out to R, reads the value and the slope there, and glues the analytic form on. All three quantities come out of that gluing:

quantity	**comes from**
a	where the asymptotic straight line crosses zero
r_0	the integral of the difference between that line and the true solution
E	the energy that makes the solution glue onto the exponential

Why: Integrating beyond R would mean computing numerically something already known exactly, and paying for it in numerical error.

5 The four potentials

The choice is not decorative: each shape stresses a different numerical hypothesis.

Potential	**Form**	**What it stresses**
Square well, Eq. (70)	$-v\mu^2$ for $r < 1/\mu$	a **discontinuity**
Pöschl-Teller, Eq. (116)	$-v\mu^2/\cosh^2(\mu r)$	**smooth**; closed form for a only
Gaussian, Eq. (120)	$-v\mu^2 e^{-\mu^2 r^2}$	tail dying **faster than exponentially**
Lennard-Jones, Eq. (121)	$\frac{1}{2}(C_{12}/r^{12} - C_6/r^6)$	**hard core** plus van der Waals tail

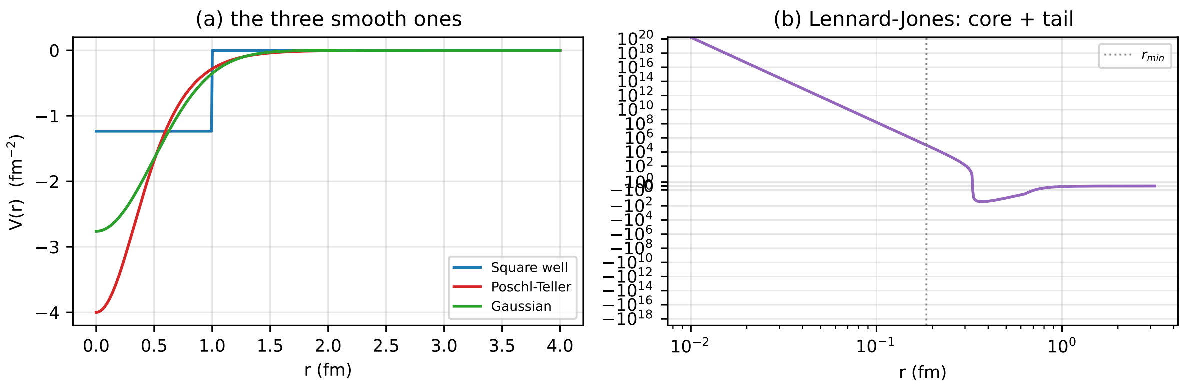

Figure 1: The four potentials, all tuned to unitarity ($|a| \to \infty$). The Lennard-Jones needs logarithmic axes because of its hard core; the dotted line marks `r_min`, the radius where V reaches `V_CORE` $= 10^5$ and integration starts. Reference [3] prescribes 10^{10} for that cut; Sec. 15 shows why 10^5 is already converged.

5.1 The range R, in closed form

Each class needs to know where its potential stops acting. **Only one of the four actually has such a radius.** The square well is the exception: it is identically zero beyond $R = 1/\mu$, so its range is a property of the potential. The other three never truly vanish — a Gaussian decays forever, a van der Waals tail falls off as a power law — and for them a finite range does not exist as a mathematical fact. What exists is a *choice*:

$$|V(R)| = \texttt{V\_ZERO} = 10^{-12}, \tag{13}$$

a truncation we impose because the integration has to stop somewhere. Calling it "the range" is shorthand, and the number 10^{-12} is a knob, not a constant of nature: it belongs in the error budget alongside `V_CORE`, which is the same kind of choice at the other end. Sec. 18.2 shows one consequence — because these thresholds are absolute while V rescales as $1/L^2$, the truncated problem is not exactly scale invariant even though the physics is.

With that understood, Eq. (13) is solved *on paper* for each shape:

$$\text{Pöschl-Teller:} \quad \frac{v\mu^2}{\cosh^2(\mu R)} = \varepsilon \quad \Longrightarrow \quad R = \frac{1}{\mu}\,\mathrm{arccosh}\sqrt{\frac{v\mu^2}{\varepsilon}} \tag{14}$$

$$\text{Gaussian:} \quad v\mu^2 e^{-\mu^2 R^2} = \varepsilon \quad \Longrightarrow \quad R = \frac{1}{\mu}\sqrt{\ln\frac{v\mu^2}{\varepsilon}} \tag{15}$$

$$\text{Lennard-Jones (tail):} \quad \tfrac{1}{2}C_6/R^6 = \varepsilon \quad \Longrightarrow \quad R = \left(\frac{C_6}{2\varepsilon}\right)^{1/6} \tag{16}$$

$$\text{Lennard-Jones (core):} \quad \tfrac{1}{2}C_{12}/r_{\text{min}}^{12} = V_{\text{core}} \quad \Longrightarrow \quad r_{\text{min}} = \left(\frac{C_{12}}{2V_{\text{core}}}\right)^{1/12} \tag{17}$$

Why: An earlier implementation searched for these radii numerically, over a global 40 000-point grid. Solving on paper removed that grid, removed a helper function, and made the result exact instead of limited by the sweep resolution.

5.2 The units of the coefficients

Since $[V_{\text{code}}] = \text{fm}^{-2}$ by Eq. (3), each term of the Lennard-Jones has to carry fm^{-2} on its own:

$$\left[\frac{C_{12}}{r^{12}}\right] = \text{fm}^{-2} \implies [C_{12}] = \text{fm}^{10}, \qquad \left[\frac{C_6}{r^6}\right] = \text{fm}^{-2} \implies [C_6] = \text{fm}^4.$$

The same two exponents reappear in Sec. 16.1 as the powers of L in the scaling rule, and that is not a coincidence: a scaling law read off the units is the same statement twice.

For the other three potentials the strength v is dimensionless and the scale μ is an inverse length, $[\mu] = \mathrm{fm}^{-1}$, so $v\mu^2$ carries the fm^{-2} by itself.

5.3 A factor of two in Eq. (121)

Equation (121) of [3] reads

$$V_{LJ}(r) = \frac{\hbar^2}{m_r}\left(\frac{C_{12}}{r^{12}} - \frac{C_6}{r^6}\right),$$

which in the units used here is $C_{12}/r^{12} - C_6/r^6$, with **no** $\frac{1}{2}$. Taken literally, it does not reproduce Table 4 of the same paper. Using the published deuteron parameters $C_{12} = 0.90485319$ and $C_6 = 6.81472$:

	a (fm)	r_0 (fm)
Table 4 of [3]	5.4	1.70
with the $\frac{1}{2}$	5.405	1.699
without it	**1.435**	1.412

Equations (70) and (120), for the well and the Gaussian, carry the same $\hbar^2/m_r$ prefactor and *do* reproduce Table 3 exactly as printed. The mismatch is therefore specific to Eq. (121); the likely reading is $2m_r$ in its denominator. The code uses the $\frac{1}{2}$.

Why: The discrepancy survives casual inspection because the result *looks* right either way — the curves have the expected shape. Only the numerical comparison against the published table exposes it.

6 Every symbol in every potential

Each potential is a formula with named constants. This section says what each name means physically, what it is called in the source, and what units it carries. Nothing here is new physics; it exists so that no symbol in this document is left undefined.

Throughout, r is the separation between the two particles and $V(r)$ the potential energy at that separation. By Eq. (3) every V below carries ℓ^{-2}, where ℓ is the length unit of that potential's own parameters: fm for the model potentials, Å for the helium ones.

6.1 The three one-scale models

$$V_{\text{well}} = -v\mu^2 \;\; (r < R), \qquad V_{\text{PT}} = \frac{-v\mu^2}{\cosh^2(\mu r)}, \qquad V_{\text{gauss}} = -v\mu^2 e^{-\mu^2 r^2}$$

symbol	what it is	in the code	units
v	**how deep**, as a pure number. Written this way so that v alone decides whether a bound state exists, independently of range.	`v`, first argument	—
μ	**how wide**, inversely: large μ is a narrow potential. Its reciprocal $1/\mu$ is the range.	`mu`, second argument	fm^{-1}
$v\mu^2$	the depth in energy units. The two always appear together because only that product has the dimensions of V.	—	fm^{-2}
R	where the potential stops acting. Exact for the well, $R = 1/\mu$; a chosen truncation for the other two, Eq. (13).	`self.R`	fm

The three differ only in shape. The well is flat then vanishes abruptly; the Pöschl-Teller falls off exponentially; the Gaussian falls off faster than any exponential. All three are purely attractive: $V < 0$ everywhere.

6.2 Lennard-Jones

$$V_{LJ}(r) = \tfrac{1}{2}\left(\frac{C_{12}}{r^{12}} - \frac{C_6}{r^6}\right)$$

symbol	what it is	in the code	units
C_{12}	strength of the **repulsion**. Two atoms cannot overlap: their electrons are forbidden to share a state, and pushing them together costs energy that rises steeply. The exponent 12 is chosen for convenience, not derived.	`C12`, *second* argument	fm^{10}
C_6	strength of the **attraction**. Even with no net charge, the electron cloud of each atom fluctuates; the fluctuations correlate and pull the atoms together. The exponent 6 *is* derived, from second-order perturbation theory on the dipole interaction.	`C6`, *first* argument	fm^4
$r_{\min}$	where integration starts, at $V =$ `V_CORE`. The repulsion diverges at $r = 0$ and cannot be integrated through.	`self.r_min`	fm

Why: The constructor takes C_6 *first*, although the potential is universally named "12–6" after the exponents in descending order. Swapping them changes nothing that raises an error, because both are positive, and the tuning still converges — onto the wrong labels.

6.3 The Aziz family: HFDHE2, HFD-B, LM2M2

$$V(r) = \varepsilon\Big[\underbrace{Ae^{-\alpha x+\beta x^2}}_{\text{repulsion}} - \underbrace{F(x)\Big(\tfrac{C_6}{x^6} + \tfrac{C_8}{x^8} + \tfrac{C_{10}}{x^{10}}\Big)}_{\text{attraction, damped}}\Big], \qquad x = \frac{r}{r_m}$$

symbol	what it is	in the code	units
ε	**depth of the well**: the energy that binds the two atoms at their preferred separation. For helium about 10.9 K, which is minute, and is why the dimer barely holds together.	`EPS`	K
r_m	the **preferred separation**, where V is deepest. About 2.96 Å. By construction $V(r_m) = -\varepsilon$, which is what the transcription test checks.	`RM`	Å
$x = r/r_m$	separation measured in units of r_m, so $x = 1$ is the bottom of the well.	—	—
A	**height** of the repulsive wall, in units of ε. Of order 10^5: the wall is five orders above the well.	`A`	—
α	**how fast** the wall falls off with distance.	`ALPHA`	—
β	a **curvature correction** to that fall-off. Zero for HFDHE2; adding it is what the "B" in HFD-B refers to.	`BETA`	—
C_6, C_8, C_{10}	the **attraction**, in decreasing order of reach. C_6 is dipole with dipole, C_8 dipole with quadrupole, C_{10} the next correction. Each is weaker and shorter-ranged than the last.	`C6`, `C8`, `C10`	—
$F(x)$	the **brake**. The $1/r^6$ law is derived for atoms far apart; close in it is wrong and diverges. F switches the attraction off smoothly.	inline in `V`	—
D	**where the brake acts**: $F = 1$ above $x = D$ and is cut off below.	`D`	—

LM2M2 adds one term to the same form,

$$V_a(x) = A_a\left[\sin\!\left(\tfrac{2\pi(x-\zeta_1)}{\zeta_2-\zeta_1} - \tfrac{\pi}{2}\right) + 1\right], \qquad \zeta_1 \le x \le \zeta_2,$$

a single smooth bump of height A_a spanning ζ_1 to ζ_2 and zero elsewhere (`AA`, `Z1`, `Z2`). Its peak is 0.0026 in units of ε — about a quarter of a percent of the well depth — and it moves the dimer energy from −1.685 to −1.303 mK, a change of 23%. That ratio is the halo amplification of Eq. (34) seen from another direction.

6.4 TTY

$$V(r) = A\big[\underbrace{D\,\chi^p e^{-2\beta\chi}}_{\text{repulsion}} - \underbrace{\textstyle\sum_{n=3}^{N} C_{2n} f_{2n}(\chi)\,\chi^{-2n}}_{\text{attraction, damped}}\big], \qquad \chi = r/a_0$$

symbol	what it is	in the code	units
A	overall **energy scale**.	`A_K`	K
χ	separation in **bohr**, not ångström. This is the only potential here whose native length is atomic, so r is converted on the way in.	`BOHR`	—
D	**strength** of the repulsive term.	`D`	—
β	**decay rate** of the repulsion.	`BETA`	bohr^{-1}
$p = \frac{7}{2\beta} - 1$	**not a free parameter.** It is fixed by how the atomic wavefunction falls off far from the nucleus, which is what makes this potential theoretical rather than fitted.	`self.p`	—
C_6, C_8, C_{10}	as above. Only these three are given; the rest follow from the published recurrence $C_{2n} = (C_{2n-2}/C_{2n-4})^3 C_{2n-6}$ up to $n = N$.	`C6`, `C8`, `C10`	—
N	how far the dispersion series runs: $N = 12$, so C_6 through C_{24}.	`N`	—
f_{2n}	the **brake**, one per term, and here it depends on distance through $b(\chi) = 2\beta - p/\chi$. It is the regularised lower incomplete gamma $P(2n+1, b\chi)$; Sec. 20 explains why it must be evaluated that way.	`gammainc`	—

7 Numerov: the integrator

Numerov solves equations of the form $u'' = W(r)\,u$, which is exactly Eq. (4) with $W = 2(V - E)$. The recursion is

$$u_{i+1} = \frac{(12 - 10f_i)u_i - f_{i-1}u_{i-1}}{f_{i+1}}, \qquad f_i = 1 - \frac{h^2 W_i}{12}. \tag{18}$$

Where it comes from

Expanding $u_{i\pm1}$ in Taylor series and adding,

$$u_{i+1} + u_{i-1} = 2u_i + h^2 u_i'' + \frac{h^4}{12} u_i^{(4)} + O(h^6).$$

A second-order method simply discards the h^4 term. Numerov does not: since $u'' = Wu$, we have $u^{(4)} = (Wu)''$, and that second derivative can be approximated by the *same* central difference, without evaluating anything at new points. Substituting and regrouping gives Eq. (18).

Why: Numerov therefore reaches order h^4 at the cost of a second-order scheme: it never evaluates W away from the grid. The price is a *three-term* recursion — each point depends on the two before it, so an error made at the start travels all the way out.

```
f = 1.0 - h * h * W / 12.0
for i in range(1, POINTS - 1):
    w[i + 1] = ((12.0 - 10.0 * f[i]) * w[i] - f[i - 1] * w[i - 1]) / f[i +
        1]
```

8 The logarithmic grid

8.1 The problem

Numerov requires $h\sqrt{|W|} \ll 1$ to represent the solution. Measured across the four potentials on a uniform grid:

potential	h (fm)	$\max\lvert W\rvert$	$h\sqrt{\lvert W\rvert}$
square well	0.00103	8.3×10^{-1}	0.001
Pöschl-Teller	0.00851	2.1×10^{0}	0.012
Gaussian	0.00401	1.6×10^{0}	0.005
Lennard-Jones	0.00626	$\mathbf{2.0 \times 10^{5}}$	**2.78**

The Lennard-Jones is squeezed from both sides: its core reaches $V = 10^5$, so $\sqrt{2V} = 447$ and the characteristic length there is 0.002 fm, while its $1/r^6$ tail pushes R out to 125 fm. **No uniform step serves both.**

The consequence was measured, and it is a real error. On a uniform grid the Lennard-Jones r_0 read 1.74257, then 1.74215, then 1.74165 as the number of points doubled: it *drifted* instead of converging, and the drift was larger than the change being resolved.

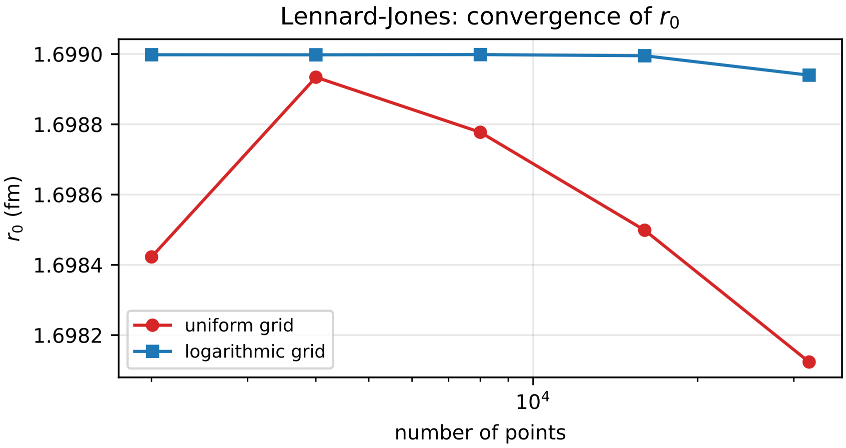

Figure 2: Convergence test. On a uniform grid (red) r_0 never settles. On a logarithmic grid (blue) it locks in at the fifth decimal. A result that moves when the grid is refined is not a result.

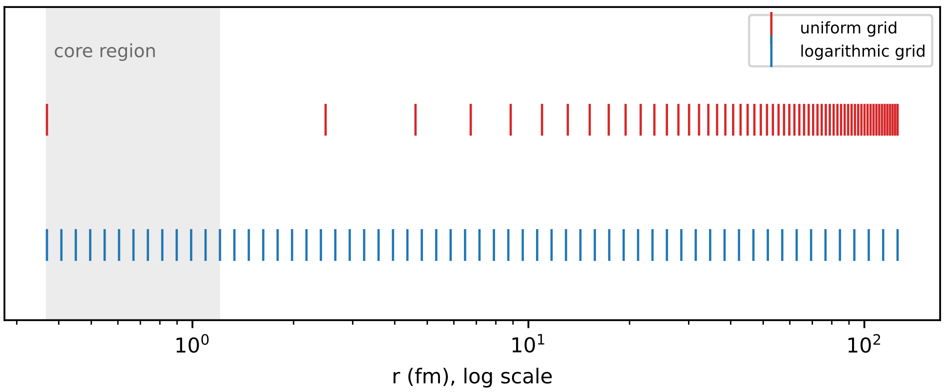

Figure 3: Point distribution, same number of points in both cases. The uniform grid wastes almost everything on the tail and leaves the core with a handful of points; the logarithmic grid does the opposite.

8.2 The calculation

Strictly the argument of a logarithm has to be dimensionless, and r is in fm. So the substitution is

$$x = \ln\left(\frac{r}{r_*}\right), \qquad r_* = 1 \text{ fm}, \tag{19}$$

and $r = r_* e^x$. Choosing $r_* = 1$ fm means every number below is unchanged — which is why the distinction is easy to drop — but the definition is only well posed with it written. Below we keep $r_* = 1$ implicit and write $r = e^x$.

With that, and $u = e^{x/2} w(x)$, using $\frac{d}{dr} = e^{-x}\frac{d}{dx}$:

$$u' = e^{-x}\frac{d}{dx}\left(e^{x/2}w\right) = e^{-x/2}\left(\tfrac{1}{2}w + w'\right), \tag{20}$$

$$u'' = e^{-x}\frac{d}{dx}\left[e^{-x/2}\left(\tfrac{1}{2}w + w'\right)\right] = e^{-3x/2}\left(w'' - \tfrac{1}{4}w\right). \tag{21}$$

Substituting into Eq. (4), $e^{-3x/2}(w'' - \frac{1}{4}w) = 2(V - E)e^{x/2}w$, so

$$\boxed{w''(x) = \left[2\left(V(e^x) - E\right)e^{2x} + \tfrac{1}{4}\right] w(x)} \tag{22}$$

Two things matter here. First, a $\frac{1}{4}$ **appears** that was not in the original equation — it comes from the substitution $u = e^{x/2}w$, not from the physics. Second, and decisively: Eq. (22) is still of the form $w'' = Ww$, which is the *only* thing Numerov requires. The recursion does not change at all; only what goes into it does.

Why: Changing the independent variable is standard when a problem has two very different scales. Here they differ by five orders of magnitude: 0.002 fm in the core against 125 fm in the tail.

```
r_start = pot.r_min if pot.r_min > 0.0 else pot.R * 1e-6
x = np.linspace(math.log(r_start), math.log(pot.R), POINTS)
h = x[1] - x[0]
r = np.exp(x)
W = 2.0 * (pot.V(r) - E) * r * r + 0.25
```

9 The two starts

Numerov needs *two* initial values, and the code picks between two cases:

```
if pot.r_min > 0.0:
    w[1] = h * math.exp(x[0] / 2.0)       # hard core: u = 0, u' = 1 there
else:
    w[0] = math.sqrt(r[0])                # regular origin: u ~ r
    w[1] = math.sqrt(r[1])
```

Regular origin (well, Pöschl-Teller, Gaussian). Near $r = 0$ with finite V, the regular solution behaves as $u \simeq r$. Since $w = u\,e^{-x/2} = u/\sqrt{r}$, that gives $w \simeq \sqrt{r}$, which is exactly what the code writes.

Hard core (Lennard-Jones). There V blows up and u is exponentially small; we impose $u(r_{\min}) = 0$ with unit slope, and the same conversion $w = u/\sqrt{r}$ gives $w_1 \simeq h\,e^{x_0/2}$.

Why: An incorrect start is not a local error: by the three-term recursion, the error propagates through the whole integration. A measurement from this work: for the Lennard-Jones, the fourth-order start correction $u_1 = h + h^3 W_0/6$ amounts to *129%* of h. It did not fix that version's problem — the cause was the grid — but it shows the scale of what is at stake.

10 The slope at the edge

Gluing the solution onto the analytic form needs $u(R)$ and $u'(R)$. The value comes out directly; the derivative uses a five-point one-sided difference:

$$\left.\frac{dw}{dx}\right|_R \simeq \frac{25w_N - 48w_{N-1} + 36w_{N-2} - 16w_{N-3} + 3w_{N-4}}{12h} + O(h^4), \tag{23}$$

and the conversion back to physical radius comes from the first line of the calculation in Sec. 8:

$$\left.\frac{du}{dr}\right|_R = e^{-x_N/2}\Big(\tfrac{1}{2}w_N + \left.\frac{dw}{dx}\right|_R\Big). \tag{24}$$

Why: The one-sided formula is used instead of a central difference for two reasons. It is order h^4, matching the rest of the method — a central one would be $O(h^2)$ and would spoil the global order. And it uses *interior points only*, so it never crosses the square well's discontinuity.

The discontinuity finding

If a Numerov step straddles the well's jump, the method assumes W is smooth across the three points it uses — and it is not. The convergence order was measured:

n	error in a	measured order
2001	8.9×10^{-4}	—
4001	4.5×10^{-4}	1.00
8001	2.2×10^{-4}	1.00
16001	1.1×10^{-4}	1.00

Order exactly 1, not 4. With the grid ending on the edge and the one-sided derivative, the same case drops to 1.9×10^{-10} — **six orders of magnitude**, with the same number of points.

11 What the code measures: a and r_0

11.1 The scattering length

Beyond R the solution is the straight line $u = C(1 - r/a)$. It crosses zero at $r = a$, and knowing $u(R)$ and $u'(R)$ fixes the line:

$$\boxed{a = R - \frac{u(R)}{u'(R)}} \tag{25}$$

Why: This is an *exact reading*, not a fit. No least squares, no extrapolation. The only source of error is the integration out to R.

Written that way, a has a pole at $u'(R) = 0$ and the code would need a branch there. It does not, because the *reciprocal* is regular:

$$\frac{1}{a} = \frac{u'(R)}{R\,u'(R) - u(R)}, \tag{26}$$

which at $u'(R) = 0$ evaluates to $0/[-u(R)] = 0$. That is what the code computes, and a is recovered by inversion:

```
inv_a = slope / (slope * pot.R - u[-1])
a = 1.0 / inv_a
```

There is no guard on that inversion, deliberately. `inv_a` vanishes only when the slope is exactly zero, which is the unreachable case above: instrumented over the twelve published parameter sets and the four measured potentials, it did not occur once. A branch for a state the program cannot reach is an untested line making a claim about the physics. And if it did occur, IEEE arithmetic already answers better than a constant would: $1/(+0) = +\infty$ and $1/(-0) = -\infty$, so the sign follows the side the slope vanished on — which is precisely the direction-dependence the limit has.

Why: That branch is mathematically correct and **numerically unreachable**. A floating-point slope is never exactly zero. Swept across the square well's resonance, it measures

v	a	$u'(R)$
1.2214	-80.3	7.9×10^{-3}
1.2325	-809	7.9×10^{-4}
1.2337	-1.82×10^{6}	3.5×10^{-7}
1.2349	$+812$	-7.8×10^{-4}
1.2460	$+81.9$	-7.8×10^{-3}

Small, then small and negative — never zero. Two things follow.

First, $a = \infty$ never actually occurs, so unitarity has to be recognised by a threshold rather than by an equality.

Second, a does not have a limit at the resonance at all: it runs to $-\infty$, is undefined, and returns from $+\infty$. Both signs are correct one-sided limits and neither is *the* answer, so any single value returned there is a placeholder rather than a result. Equation (26) is what makes this harmless — $1/a$ passes through zero without noticing, and it is $1/a$ that the tuning chases and $|r_0/a|$ that the node count tests. The placeholder cannot propagate, and `test_unitarity_sign_does_not_leak` asserts as much by flipping it.

What "infinite" means, numerically

The question is not rhetorical: at what value do we call a infinite? Asked about a itself the question has no answer, because a carries a length and the answer would then depend on the system. Asked about a dimensionless ratio it does. The right ratio is r_0/a: it is invariant under the rescaling of Sec. 16.1, and it vanishes at unitarity by construction. It is also the quantity Sec. 18.3 argues should be reported anyway.

R/a will not serve. R is a truncation we chose, not physics — and for the Lennard-Jones at unitarity $R/a = 1.5 \times 10^{-3}$, which is not small.

Measured over the twelve published cases:

	$\lvert r_0/a\rvert$
at unitarity	2.2×10^{-11} to 2.1×10^{-5}
where an exterior node must count	3.14×10^{-1}

a gap of about 1.5×10^4, so

$$\texttt{UNITARITY} = 10^{-3} \tag{27}$$

sits 47× above one group and 314× below the other. Every one of the twelve cases gets the Table 2 count for any threshold from 10^{-2} to 10^{-4}; it fails at 10^{-5}, where the Lennard-Jones at unitarity stops being recognised. Three decades of margin, and all three are covered by a test. A threshold that worked at only one value would be a fit, not a criterion.

Why: The reading must be taken *outside* the range, not at the last point where the potential still acts. Inside a square well $V \neq 0$, so the solution there is not the straight line and the tangent construction does not apply. Measured on the deuteron well, matching one point too early biases the bound state by 0.44% — four orders above the 10^{-10} the method otherwise reaches.

11.2 The normalisation, and why it is needed

Equation (4) is homogeneous: if u is a solution, so is Cu. The start chose an arbitrary normalisation, so u is not yet comparable with anything. We fix it by requiring that it meet the line at the edge:

$$u \longleftarrow u\,\frac{1-R/a}{u(R)} \qquad \Longrightarrow \qquad u(R)=g_0(R), \quad g_0(r)\equiv 1-\frac{r}{a}. \tag{28}$$

Why: Without this the effective-range integral would not be defined — it compares u against g_0, and the comparison only makes sense once both are on the same scale.

11.3 The effective range

$$r_0 = 2\int_0^\infty \Big[g_0(r)^2 - u(r)^2\Big]\,dr. \tag{29}$$

The integrand measures *how far the true solution departs from the line.* Outside the range $u = g_0$ and it vanishes identically, so the integral does end at R.

Why: It is inaccurate to read r_0 as "the size of the potential". For the square well the temptation is strong, because at every pole of a we get $r_0/R = 1$ exactly — and it is precisely because the analogy works so well there that it should not be generalised. In general r_0 is not a geometric radius:

- it can be **negative**;
- it depends on the shape of the *interior* wavefunction, not on where the potential stops;
- it can grow large near special structures, without the potential growing at all;
- it need not coincide with any radius one could draw on a plot of V.

What Eq. (29) says is narrower and safer: r_0 measures the failure of the true solution to be the straight line, weighted over all r. That is a statement about the wavefunction, and only indirectly about the potential.

Two details in the code:

The factor r. On the logarithmic grid $dr = r\,dx$, so the integral becomes $\int(\dots)\,r\,dx$. That is where the `* r` in the Simpson call comes from.

The analytic piece. The grid starts at r_{start}, not at zero. The stretch $[0, r_{\text{start}}]$ is added by hand: there u is either zero (inside a hard core) or proportional to r, and in both cases its contribution is $O(r_{\text{start}}^3)$ while the line's is $O(r_{\text{start}})$, leaving

$$2\int_0^s g_0^2\,dr = 2\left(s - \frac{s^2}{a} + \frac{s^3}{3a^2}\right), \qquad s \equiv r_{\text{start}}. \tag{30}$$

Why: This term used to apply only to the Lennard-Jones. Measurement showed the square well carried a residual error of 2.4×10^{-6} in r_0 that was exactly this missing piece. Making the term unconditional removed an `if` *and* improved precision by three orders of magnitude — down to 5.6×10^{-12}.

```
r0 = 2.0 * simpson((line ** 2 - u ** 2) * r, x=x)
s = r_start
r0 = r0 + 2.0 * (s - s ** 2 / a + s ** 3 / (3.0 * a ** 2))
```

12 Counting bound states

The code counts *zeros of the zero-energy solution.* It is a striking fact that the scattering solution, at $E = 0$, knows how many bound states the potential has.

Reference [3] treats this as a mandatory check rather than as a result: *"checking the number of nodes of the radial function is necessary to guarantee that it is indeed the situation we wanted to reproduce"*, expecting a nodeless $u(r)$ for $a < 0$ and one node for $a > 0$.

There is a subtlety. Beyond the edge the solution is the line $g_0 = 1 - r/a$, which crosses zero at $r = a$. If $a > R$ that zero is physical and counts. If a is effectively infinite, the zero sits at infinity and does not — hence the test

```
    if pot.R < a and abs(r0 / a) > UNITARITY:
```

with `UNITARITY` fixed in Sec. 11.1. An earlier version read `R < a < 1e4`, a bound with a length hidden in it and no argument behind the number.

```
    nodes = 0
    for i in range(1, POINTS - 1):
        if u[i] * u[i + 1] < 0.0:
            nodes = nodes + 1
    if pot.R < a < 1e4:
        nodes = nodes + 1
```

Why: The count is not optional. Two solutions can share (a, r_0) and differ in node count — and then they are *not* the same physical state. Every tuning ends by checking it.

A claim that used to live here, and is wrong: that the loop starts at `i = 1` because $u_0 = 0$ would otherwise count as a sign change. The test is `u[i]*u[i+1] < 0.0`, and a product of zero is not less than zero, so it never could. Running from `i = 0` and from `i = 1` gives the same count on all four potentials, and in three of them u_0 is not zero anyway. What did fix the original symptom is not recorded and is not reconstructed here.

13 The bound state

Beyond the edge, with $E < 0$, the solution must be the decaying exponential. That gives the matching condition

$$\boxed{g(E) \equiv \frac{u'(R) + \kappa\, u(R)}{\max\bigl(|u(R)|,\, |u'(R)|,\, 1\bigr)} = 0, \qquad \kappa = \sqrt{-2E}} \tag{31}$$

Why: Two separate decisions are packed into that expression.

The sum, not the ratio $u'/u + \kappa$. Measured: with the ratio, `brentq` fails to bracket for three of the four potentials — it reports that the two ends of the interval carry the same sign. The sum brackets for all four. (An earlier version of these notes justified this by saying that $u(R)$ passes through zero because "the deuteron has a node". That is wrong twice over. The deuteron's bound state is the ground state and has *no* node — what carries a node is the zero-energy scattering solution of a potential that supports a bound state, which is a different function. And $|u(R)|$ never comes near zero here: swept across the bracket, its smallest value over the four potentials is 7×10^{-3}.)

The denominator. It is normalisation, and it does not move the root — removing it changes the answer by 3×10^{-16}. What it fixes is scale. The raw residual $u'(R) + \kappa u(R)$ is of order 10^{-2} for the square well and 10^{21} for the Lennard-Jones, because u at the edge spans that range after climbing out of the hard core. A single absolute tolerance cannot serve both. Dividing brings every potential to order unity.

Why there is no search

The finite-range formula already delivers the answer to about 1%. It is the guess, and `brentq` merely refines it within $[1.8\,g,\ 0.3\,g]$:

```
    return brentq(gap, 1.8 * guess, 0.3 * guess, xtol=1e-15)
```

Why: An earlier implementation scanned 120 energies on a logarithmic grid to find where the function changed sign. That was waste: the approximate answer was already in hand and was being ignored. Replacing the scan by the guess made the calculation 3.5× faster *and* more accurate. There is also a conceptual bonus: the bracket always working is itself a statement that the effective-range expansion is good.

14 The inverse problem: tuning

Given a target (a, r_0), find the parameters. It is a nonlinear 2×2 system, and a has *poles*: it jumps from $+\infty$ to $-\infty$ every time a new bound state appears.

14.1 Two nested loops

`tune`	outer loop: moves the **scale** until r_0 matches
`match_a`	inner loop: moves the **strength** until $1/a$ matches
`find_root`	widens a bracket until the sign flips, then bisects

Each loop solves *one* equation in *one* variable. Inside the outer loop, `r0_error` runs the entire inner loop again for every scale it tries — which is why this is the slow part, about 35 of the 54 seconds.

The decision that makes it work: chase $1/a$

Why: a has poles; $1/a$ is smooth and crosses zero. Bisection *dies* on a pole and thrives on a zero. As a bonus, unitarity becomes $1/a = 0$ and needs no special case at all.

```
    if math.isinf(a_target):
        inv_a_target = 0.0
    else:
        inv_a_target = 1.0 / a_target
```

Why not a 2D Newton

The level sets of $1/a$ and r_0 in the (strength, scale) plane are *nearly parallel* over much of the domain. A Newton solver on both equations at once would meet an ill-conditioned Jacobian exactly there, and take large steps in the wrong direction. Nested bisection cannot diverge once the sign is bracketed — slower per iteration and far more reliable for code someone else will run.

The six return values

`tune` returns `(strength, scale, a, r0, nodes, ok)`. The last is a flag: false if r_0 failed to converge *or* if the node count missed its target. Without it, a tuning that landed on the wrong state would pass unnoticed.

Why: What comes out of `tune` is a different pair of numbers for every potential — there is no shared parameter. What is shared is the pair (a, r_0), and with it everything a low-energy measurement can see. That is the point of Sec. 1.

15 Every constant, one by one

No number in the code is a guess. This table is the justification for each.

constant	value	where	justification
`V_ZERO`	10^{-12}	defines R	Below this the potential is treated as gone. Measured: cutting at 10^{-8} truncates the Lennard-Jones tail and misses r_0 by 0.16%; 10^{-12} closes to 0.004%.
`V_CORE`	10^5	LJ start	Where integration begins inside the core. Ref. [3] prescribes $V(r_{\min}) \sim 10^{10}$; we measured that 10^3 breaks the physics (r_0 off by 0.2%) while anything from 10^4 up changes nothing. Starting deeper only costs stiffness.
`POINTS`	8001	grid size	Total error is truncation (falls with n) plus roundoff (grows with n); 8001 sits at the minimum. The three purely attractive potentials are at 10^{-11} and do not care. The Lennard-Jones sets the value: past ∼8000 its r_0 *degrades* again — 3×10^{-5} at 32001, 3×10^{-4} at 64001.
`r_start`	$R \times 10^{-6}$	regular origin	Where the logarithmic grid begins when there is no core. It cannot be 0 because $\ln 0$ does not exist. The stretch below it enters analytically.
$[1.8g,\ 0.3g]$	—	`brentq`	Bracket around the finite-range guess. It works on both systems tested; that it works at all is a statement about the quality of the expansion.
1.3	—	`find_root`	Bracket widening factor, up to 40 attempts. If no sign change is found it raises rather than returning garbage.
10^{-7}, 10^{-6}	—	tolerances	Convergence of r_0 in `tune` and in the tests. Measured: the tuning hits its target to 10^{-10}, comfortably inside them.

16 The tabulated data

Five dictionaries, all traceable:

TARGETS	Table 2 of [3]: the three targets. The bound-state count is *not* a column of that table; it comes from the text of Sec. 4.5.
PUBLISHED	Tables 3 and 4: the published parameters. Used as the initial guess *and* as the validation target.
SYSTEMS	Table 1: two real systems, with measured energies.
GUESS	The deuteron solution, used as a starting point for other systems.
SOURCES	The three references with DOI, plus the D1 divergence.

16.1 Scale invariance in GUESS

The problem is scale invariant: multiplying all lengths by L takes $(a, r_0) \to (La, Lr_0)$. So attacking another system only needs the guess rescaled — but the exponents follow from the units, not from the powers of r alone. Since $[V_{\text{code}}] = \text{fm}^{-2}$, the rescaled potential must satisfy $V'(Lr) = V(r)/L^2$. For the repulsive term,

$$\frac{C'_{12}}{(Lr)^{12}} = \frac{1}{L^2}\frac{C_{12}}{r^{12}} \quad \Longrightarrow \quad C'_{12} = L^{10}\, C_{12},$$

and the same argument on the attractive term gives $C'_6 = L^4 C_6$. So

$$\mu \to \frac{\mu}{L}, \qquad C_6 \to C_6\, L^4, \qquad C_{12} \to C_{12}\, L^{10}. \tag{32}$$

These are exactly $[C_6] = \text{fm}^4$ and $[C_{12}] = \text{fm}^{10}$ read backwards. An earlier version of these notes carried $C_{12} \to C_{12} L^6$, which comes from treating V as dimensionless; it is wrong, and it misses C_6 entirely. The consequence was not cosmetic — see Sec. 18.2.

16.2 The constant that is tabulated nowhere

$\hbar^2/2m_r$ appears in no table. We obtain it by inverting the zero-range formula on the published pair (a, E_{zr}):

$$E_{zr} = -\frac{\hbar^2/2m_r}{a^2} \quad \Longrightarrow \quad \frac{\hbar^2}{2m_r} = -E_{zr}\, a^2. \tag{33}$$

system	inferred	literature	deviation
deuteron	41.462 MeV·fm^2	41.47	0.02%
^{4}He dimer	12.095 K·Å^2	12.12	0.21%

Why: This is a test, not a convenience. Recovering both known constants proves Table 1 is internally consistent — and, incidentally, that the factor of 2 in Eq. (5) is right.

The three tunings that do not match, and why

Nine of the twelve tunings land within 0.2% of the published parameters. Three do not, and they do not fail for the same reason.

Two of them are the D1 divergence. (nn, Lennard-Jones) at 2.68% and (deuteron, Pöschl-Teller) at 2.65%. The cause is *not* numerical: refining the grid moves the deviation from 2.70% to 2.69%. Table 3 of [3] reports $r_0 = 1.73$ for the mPT deuteron row and Table 4 reports $r_0 = 2.71$ for the Lennard-Jones nn row, while Table 2 of the same paper lists the targets as 1.70 and 2.70. Feeding the published parameters back through the solver confirms it: they produce $r_0 = 1.7300$ and 2.7080, not the targets. We tune to the target and land elsewhere — correctly.

The third is a flat direction, not a disagreement. (deuteron, Lennard-Jones) sits 1.13% away, but here the published parameters *do* reproduce the target: they give $r_0 = 1.6990$ against 1.7000, an error of 0.06%. Both parameter pairs are therefore valid. What happens is that the two Lennard-Jones coefficients are strongly correlated. Starting from the published pair and moving to ours:

change	effect on r_0
C_6 alone, +0.47%	−1.09%
C_{12} alone, +1.13%	+1.18%
both together	+**0.05**%

Each coefficient on its own moves r_0 by about 1%; together they cancel. The inversion $(a, r_0) \to (C_{12}, C_6)$ has a nearly flat direction, so the observable is pinned down far more tightly than the parameters that produce it. This is worth stating plainly because the naive reading — "the parameters disagree by 1%, so something is wrong" — is exactly backwards. Note also that Lennard-Jones is the *most* sensitive of the four potentials ($\mathrm{d}\ln r_0/\mathrm{d}\ln C_6 = 2.28$, against 0.42 for the well), so the flatness comes from the correlation between the two coefficients, not from any insensitivity of the potential.

General rule: a discrepancy that survives refinement is never numerical.

17 The checks

`test_lab.py` runs eight checks, and every one compares against something the code did **not** compute:

check	**external anchor**
a of the well	Eq. (80) of [3], closed form
r_0 of the well	Eq. (92) of [3], closed form
$r_0/R = 1$ at the poles	exact prediction, caption of Fig. 6 of [3]
bound state	analytic condition $k\cot(kR) = -\kappa$
node counts	Table 2, all 12 cases
tuning	Tables 3 and 4, all 12 cases
inferred constants	41.47 and 12.12 from the literature
grid convergence	halving the points may change nothing

Why: No check compares against a stored previous run. Such a check would only prove the code is consistent with its own mistakes.

The last one is what *found* the uniform-grid bug. It halves the points rather than doubling them, deliberately: since the total error has a minimum, the useful question is "have we converged coming from below?" and not "what happens deep in the roundoff regime?".

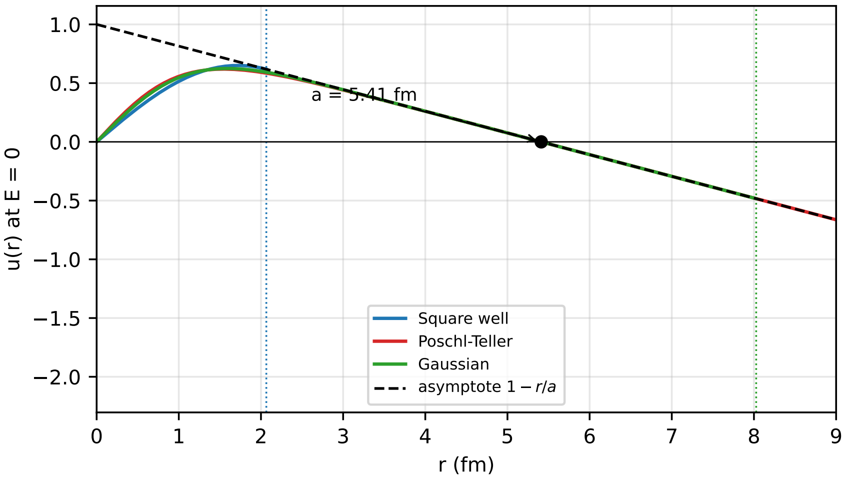

Figure 4: Three potentials tuned to the same (a, r_0) of the deuteron: the wavefunctions are *identical outside* the range and *completely different inside*. Dotted lines mark where each potential ends. Everything a low-energy experiment can see lives outside.

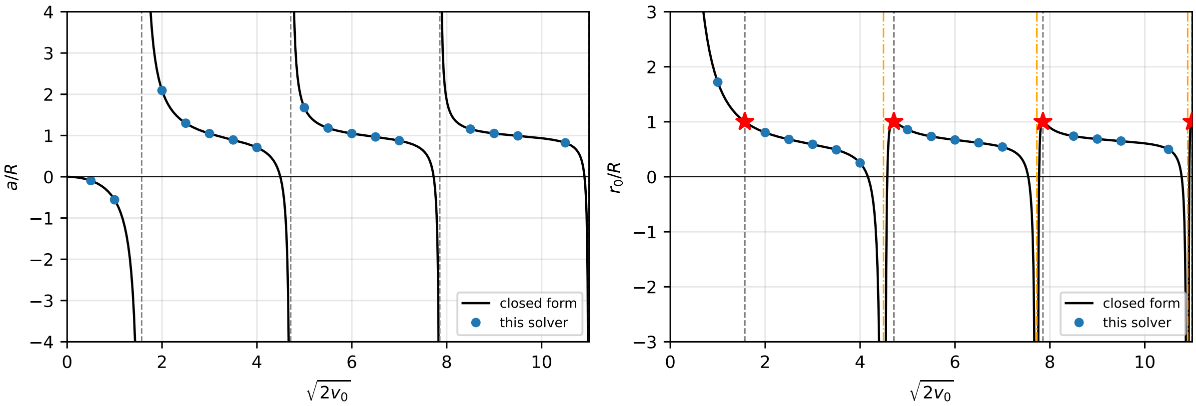

Figure 5: Reproduction of Figs. 5 and 6 of [3]. Left: a/R diverges at $\sqrt{2v_0} = \pi/2 + n\pi$, where a new bound state appears. Right: at each of those points $r_0/R = 1$ *exactly* (red stars), as stated in the caption of Fig. 6 of [3]. Orange lines mark the *other* family of singularities, the roots of $\tan x = x$, where $a = 0$ and it is r_0 that diverges.

18 Results

18.1 Binding energies

	deuteron ($r_0/\lvert a\rvert = 0.32$)		^{4}**He dimer** ($r_0/\lvert a\rvert = 0.089$)	
method	E (MeV)	deviation	E (K)	deviation
experiment	−2.224	—	-1.620×10^{-3}	—
zero range	−1.416	−36.3%	-1.480×10^{-3}	−8.6%
finite range	−2.223	−0.05%	-1.628×10^{-3}	+0.47%
square well	−2.204	−0.91%	-1.627×10^{-3}	+0.46%
Pöschl-Teller	−2.227	+0.11%	-1.628×10^{-3}	+0.47%
Gaussian	−2.214	−0.47%	-1.627×10^{-3}	+0.46%
Lennard-Jones	−2.203	−0.92%	—	—

Two systems separated by 13 orders of magnitude in energy, described by the same two numbers and the same two formulas. That is the universality of Sec. 1, measured.

The residual spread of about 1% *among* the potentials is the shape dependence — the $O(k^4)$ terms, the only thing still distinguishing a square well from a Lennard-Jones once a and r_0 are fixed.

18.2 The helium Lennard-Jones, and what the wrong exponent cost

The scaling rule of Sec. 16.1 is not decoration: it supplies the starting guess that the tuning begins from. With the wrong exponent the guess for the helium dimer was off by a factor of 443 in *both* coefficients, and — worse — it landed on the branch with **zero** bound states when the target has one:

starting guess	C_6	C_{12}	what it gives
$C_{12}L^6$, C_6 untouched	7.84	1.19×10^4	$a = 1.76$, $r_0 = 1.33$, 0 nodes
C_6L^4, $C_{12}L^{10}$	3476	5.27×10^6	$a = 24.8$, $r_0 = 7.995$, 1 node
target			$a = 90.4$, $r_0 = 8.0$, 1 node

The correct rule arrives with r_0 already within 0.06% of the target and on the right branch. The wrong one does not converge at all: it was left running for 170 s without finishing, while the corrected guess converges in 83 s.

That non-convergence is the reason the Lennard-Jones was quietly absent from the helium row — three potentials were run there, not four, and no note in these notes said so. With the exponents fixed the case runs, and it belongs with the others:

$$E(^4\mathrm{He}_2,\ \text{Lennard-Jones}) = -1.627\ \text{mK}, \qquad \text{measured}\ -1.620\ \text{mK}.$$

The lesson is about the check, not the exponent. Scale invariance is a *property* the code must satisfy, so it can be tested directly: rescale every length by L, and (a, r_0) must come back as (La, Lr_0). That test is three lines, needs no reference value, and would have caught this immediately. It is now in `test_lab.py`.

18.3 Accuracy achieved

check	error	kind
a of the well against Eq. (80)	1.9×10^{-10}	relative
r_0 of the well against Eq. (92)	5.6×10^{-12}	relative
bound state against $k\cot(kR) = -\kappa$	3.4×10^{-10}	relative
r_0/R at the four poles	1.0000000000	exact value, not an error
solver against closed forms (16 points)	8.4×10^{-9}	relative
tuning, a against target	10^{-6}	relative
tuning, r_0 against target	10^{-6}	**absolute**

Why: The last two rows were previously one row reading "10^{-10}". Splitting them is not pedantry: the code asserts `abs(r0 - target) < 1e-6`, an absolute tolerance in fm, and `abs(a/target - 1) < 1e-6`, a relative one. Those are different requirements, and at unitarity, where $a \to \infty$, only the absolute one is even meaningful. Stating "the error is 10^{-6}" without saying which is an invitation to compare two numbers that are not comparable.

Convention for the rest of this document: every error is relative unless it is marked absolute.

The dimensionless number worth reporting

Absolute and relative are the minimum. Better, where the physics offers one, is a ratio built from the problem itself. Here the natural one is r_0/a: it is dimensionless, it is invariant under the rescaling of Sec. 16.1, and it measures how close a system sits to unitarity, where it vanishes.

system	a	r_0	r_0/a	zero-range error
unitarity	∞	1.0	0	—
^{4}He dimer	90.4	8.0	0.089	8.6%
neutron-neutron	−18.5	2.7	−0.146	—
deuteron	5.4112	1.744	0.322	36.3%

Why: This column accounts for a result that would otherwise appear arbitrary. The zero-range formula misses the deuteron by 36% and the helium dimer by only 8.6% — a factor of 4.2. The ratio r_0/a differs between them by a factor of 3.6. Zero range is the approximation that throws r_0 away, so its error is governed by exactly this number, and the two factors agreeing to within 20% is the expansion behaving as advertised. Note also that r_0/a is negative for the neutron-neutron system, which is a sign that no dimensionless "size" reading of r_0 survives: see Sec. 11.

19 The error budget

Every number in this document rests on six choices that are ours, not the physics'. This section varies each one and reports what r_0 does. All entries are **relative** errors against the value at the setting actually used, and the deuteron parameters of Table 3 are the test case throughout.

`POINTS` — discretisation and round-off

`POINTS`	well	Pöschl-Teller	Gaussian	Lennard-Jones
2001	2.1×10^{-9}	1.6×10^{-10}	8.3×10^{-11}	2.6×10^{-7}
4001	1.3×10^{-10}	2.0×10^{-12}	3.9×10^{-12}	3.7×10^{-7}
8001	—	—	—	—
16001	8.2×10^{-12}	5.0×10^{-11}	1.9×10^{-11}	1.9×10^{-6}
32001	1.4×10^{-11}	2.7×10^{-10}	1.3×10^{-10}	$\mathbf{3.4 \times 10^{-5}}$

Why: This is the one parameter for which larger values are not better. The three purely attractive potentials are flat from 4001 upward — they sit at machine precision and stop caring. The Lennard-Jones does not: its error falls to 8001 and then *climbs* by two orders of magnitude out to 32001. That is round-off overtaking truncation, and it is the reason 8001 is written down rather than "as many as we can afford". The review asked for values beyond 4001 and 8001; the answer is that they are worse.

`V_ZERO` — truncating the tail

`V_ZERO`	well	Pöschl-Teller	Gaussian	Lennard-Jones
10^{-9}	0	1.4×10^{-8}	1.6×10^{-9}	$\mathbf{1.1 \times 10^{-3}}$
10^{-10}	0	1.5×10^{-9}	2.1×10^{-10}	6.4×10^{-4}
$\mathbf{10^{-12}}$	—	—	—	—
10^{-14}	0	3.2×10^{-10}	4.7×10^{-11}	3.6×10^{-4}

Why: The well is exactly zero in every row, which is the point made under Eq. (13): it is the only potential with a real range, so truncation cannot touch it. Everything else pays, and the Lennard-Jones pays 10^3 to 10^6 times more than the others, because a $1/r^6$ tail is still there when a Gaussian has long since vanished. Note that 10^{-14} is also worse than 10^{-12}: pushing the cut further out lengthens the grid and buys round-off.

`V_CORE` — a wall instead of the core

`V_CORE`	well	Pöschl-Teller	Gaussian	Lennard-Jones
10^3	0	0	0	$\mathbf{2.4 \times 10^{-3}}$
10^4	0	0	0	2.9×10^{-7}
$\mathbf{10^5}$	—	—	—	—
10^6	0	0	0	6.7×10^{-7}
10^7	0	0	0	3.2×10^{-7}

Why: Only the Lennard-Jones has a core, so only it responds — the zeros in the other three columns are exact, not rounded. From 10^4 upward the answer is converged to 10^{-6}, and 10^3 is visibly wrong. This is the measurement behind using 10^5 where [3] prescribes 10^{10}: the extra five decades buy nothing and cost stiffness.

`R_START_FRACTION` — skipping the origin

fraction of R	well	Pöschl-Teller	Gaussian	Lennard-Jones
10^{-3}	2.9×10^{-9}	$\mathbf{6.8 \times 10^{-6}}$	4.5×10^{-7}	0
10^{-4}	1.8×10^{-12}	6.7×10^{-9}	4.5×10^{-10}	0
$\mathbf{10^{-6}}$	—	—	—	—
10^{-8}	8.5×10^{-12}	1.5×10^{-11}	9.6×10^{-12}	0
10^{-10}	5.7×10^{-11}	1.1×10^{-11}	4.4×10^{-12}	0

Why: The Lennard-Jones column is exactly zero everywhere because it never uses this knob: it has a hard core, so its start comes from `V_CORE`. For the other three the piece below r_{start} is added analytically, which is why even 10^{-3} costs only 10^{-6}. Converged by 10^{-4}; the value used has four decades of margin.

This constant used to be written inline inside `numerov`. It was lifted out to module level while assembling this table — a knob that cannot be turned cannot be audited.

The edge derivative — h^4 against h^2

Replacing the five-point one-sided stencil by the three-point one, and reading the change in a:

potential	change in a
square well	3.7×10^{-6}
Pöschl-Teller	3.5×10^{-7}
Gaussian	9.2×10^{-8}
Lennard-Jones	5.0×10^{-7}

Why: The square well is hit ten times harder than the others, and that is not an accident: it is the one with a discontinuity at the edge, so the stencil is sampling a function whose derivative jumps. Dropping to $O(h^2)$ would put the matching error at 10^{-6}, four orders above the 1.9×10^{-10} that Sec. 17 reports against Eq. (80). The whole accuracy of this laboratory rests on those two extra stencil points.

`brentq` — locating the root

Deuteron binding energy, square well, as the tolerance is loosened:

`xtol`	E	change
10^{-6}	-0.02628469	4.0×10^{-6}
10^{-9}	-0.02628459	2.3×10^{-10}
10^{-12}	-0.026284587	3.7×10^{-12}
$\mathbf{10^{-15}}$	-0.026284587	—

Why: Cheap: the root finder is not where the error lives. Even 10^{-9} would leave the answer good to 10^{-10}, which is below every other term in this budget. `xtol` is set to 10^{-15} because there is no reason not to.

What the budget says

source	dominant for	size at the settings used
`V_ZERO`	Lennard-Jones	$\lesssim 10^{-4}$ if loosened
`POINTS`	Lennard-Jones	10^{-5} if pushed to 32001
edge derivative	square well	10^{-6} if dropped to h^2
`V_CORE`	Lennard-Jones	10^{-7}
`R_START_FRACTION`	Pöschl-Teller	10^{-11}
`brentq`	—	10^{-12}

Three things are worth taking away. The Lennard-Jones dominates four of the six rows, because a hard core and a power-law tail stress every truncation at once. The square well is immune to the truncation knobs and uniquely sensitive to the edge derivative, for the same reason in both cases: it is the only potential with a genuine boundary. And no single knob is close to limiting the others — which is what one wants from a budget, and is why the review's judgement that a full study "is not the case here" is fair. What is written above is the map, not the survey.

20 A potential with no free parameters

Everything so far has been a shape with two knobs, turned until it sat at a chosen (a, r_0). The Aziz HFD-B potential [2] is not that. Its parameters are fixed by spectroscopy and *ab initio* theory, so feeding it in and reading (a, r_0) out is a **prediction**. It is the only falsifiable object in this document.

$$V(r) = \varepsilon\left[Ae^{-\alpha x+\beta x^2} - F(x)\left(\frac{C_6}{x^6} + \frac{C_8}{x^8} + \frac{C_{10}}{x^{10}}\right)\right], \qquad x = \frac{r}{r_m},$$

with $F(x) = e^{-(D/x-1)^2}$ for $x < D$ and 1 otherwise.

Two things it forced

The units had to be used, not assumed. The parameters are published in kelvin and ångström. Getting them into the solver means $V_{\text{code}} = V_{\text{phys}}/(2\,\hbar^2/2m_r)$ from Eq. (6), with $\hbar^2/2m_r = 12.09$ K·Å^2; lengths then come out in ångström and V_{code} in Å^{-2}. This is the first place where the convention of Sec. 2 does work rather than sit there, and it checks out: the potential reproduces its own tabulated minimum, $V(r_m) = -\varepsilon = -10.948$ K, to six digits.

The core is soft. The Lennard-Jones diverges as r^{-12}, so the code starts outside it at $V =$ `V_CORE`. Aziz does not diverge. As $r \to 0$ the exponential saturates and $F(x)$ switches the dispersion terms off, leaving a ceiling

$$V_{\max} = \frac{\varepsilon A}{2\,\hbar^2/2m_r} = 8.35 \times 10^4,$$

which is *below* `V_CORE` $= 10^5$. There is no radius where $V =$ `V_CORE`, so $r_{\min} = 0$ and the regular-origin start applies. Asking for the hard-core radius raises "`f(a) and f(b) must have different signs`" — the bracket reporting, correctly, that the thing being looked for does not exist.

What it predicts, and what that checks

	this work	published HFD-B [1]	deviation
a	88.434 Å	88.430 Å	0.005%
r_0	7.276 Å	7.276 Å	0.00%
E	−1.692 mK	−1.69 mK	0.1%
nodes	1	1	—

This is the strongest check in this document. The four tunable potentials are fitted to (a, r_0), so reproducing (a, r_0) with them is circular. Aziz is not fitted to anything here, so agreeing to four digits with an independent calculation using the same potential tests the integrator, the matching, the effective-range integral, the bound-state search and the unit conversion simultaneously, against a number this work had no part in choosing.

Note also that Table 1 of [3] quotes 90.4 Å and −1.62 mK. Those are the Cencek *et al.* values [5], a 2012 *ab initio* surface, not HFD-B. The 2% gap between the two columns is a real difference between two potentials, and both are reproduced by the same solver.

The constant matters here, and only here

Getting that agreement required using $\hbar^2/2m_r = 12.1193$ K·Å^2 from CODATA rather than the 12.0948 that Sec. 16 *infers* by inverting the zero-range formula. The two differ by 0.20%. With the inferred value, a comes out 82.98 Å: 6% low.

That factor of thirty amplification is the physics, not a defect:

$$\frac{\mathrm{d}\ln a}{\mathrm{d}\ln(\hbar^2/2m_r)} = 48.7 \tag{34}$$

near unitarity. The helium dimer is a halo state — $|a|/r_0 = 12$ — which is another way of saying it sits close to a pole of a, and everything near a pole is amplified.

Why: Two consequences worth carrying forward. **First**, this is why the inferred constant is good enough everywhere else: the four tunable potentials are fitted to a target, so a small error in the constant is absorbed into the fitted parameters and never shows. Only a potential with no free parameters has nowhere to hide it. **Second**, Eq. (34) is a warning label for anything computed near unitarity: if 0.2% in a constant becomes 6% in a, the inputs need more digits than feels reasonable.

21 What is missing

No error bars. No number here carries an estimated discretisation uncertainty.

The Aziz HFD-B potential is now implemented — see Sec. 20. What is still missing there is a published HFD-B result to check against; see that section for what could and could not be verified.

No variable phase method. An independent second route to a, which would give a cross-check on the least reliable number here.

No finite-k phase shift. Everything is done at $E = 0$, with r_0 from the integral. Computing $\delta_0(k)$ and fitting Eq. (1) would be an independent validation — and is the test that would settle the Lennard-Jones r_0, reliable only to $\sim 10^{-6}$ and limited by roundoff.

***s*-wave only,** valid while $kR \ll 1$.

References

[1] P. Stipanović, L. Vranješ Markić and J. Boronat, *Elusive structure of helium trimers*, J. Phys. B **49**, 185101 (2016), arXiv:1607.05872. Table 1, row HFDB.

[2] R. A. Aziz, F. R. W. McCourt and C. C. K. Wong, *A new determination of the ground state interatomic potential for He_2*, Molec. Phys. **61**, 1487 (1987). Parameters as tabulated in J. Boronat and J. Casulleras, arXiv:cond-mat/9309015, Appendix A.

[3] M. Macêdo-Lima and L. Madeira, *Scattering length and effective range of microscopic two-body potentials*, Rev. Bras. Ensino Fís. **45**, e20230079 (2023). doi:10.1590/1806-9126-RBEF-2023-0079

[4] R. W. Hackenburg, *Low-energy neutron-proton effective range parameters*, Phys. Rev. C **73**, 044002 (2006). doi:10.1103/PhysRevC.73.044002

[5] W. Cencek *et al.*, *Effects of adiabatic, relativistic, and QED corrections on the pair potential of helium*, J. Chem. Phys. **136**, 224303 (2012). doi:10.1063/1.4712218

A Where each piece of the code is derived

The source is annotated line by line and is not reproduced here. This table maps each construct to the section that establishes it.

function	step	derived in
`numerov`	choice of r_{start}	Sec. 9
	grid uniform in $\ln(r/r_*)$	Sec. 8, Eq. (19)
	effective potential W	Eq. (22)
	the three-term recursion	Sec. 7
	grid ending exactly on R	Sec. 10
	five-point one-sided slope	Sec. 10
`scattering`	a from the tangent at R	Sec. 11.1
	when a counts as infinite	Sec. 11.1
	rescaling u onto the line	Sec. 11.2
	the r_0 integral, Jacobian r	Eq. (29)
	the analytic piece below r_{start}	Sec. 11.3
	counting nodes, interior + exterior	Sec. 12
`bound_energy`	the matching condition	Eq. (31)
	sum rather than ratio	Sec. 13
	the normalising denominator	Sec. 13
	why no search is needed	Sec. 13
`tune`	two nested one-dimensional roots	Sec. 14
	chasing $1/a$ and not a	Sec. 14
	the node-count guard	Sec. 12

B The data, with values

`PUBLISHED` — Tables 3 and 4, against what the tuning returns

Throughout, p_1 is the **strength** parameter and p_2 the **scale** parameter, in that order — which is the order the constructors take them in. Concretely:

potential	p_1 (strength)	p_2 (scale)
square well	v, dimensionless	μ, fm^{-1}
Pöschl-Teller	v, dimensionless	μ, fm^{-1}
Gaussian	v, dimensionless	μ, fm^{-1}
Lennard-Jones	$\boldsymbol{C_6}$, fm^4	$\boldsymbol{C_{12}}$, fm^{10}

Why: The Lennard-Jones row is worth pausing on: `LennardJones` takes C_6 *first*, even though the potential is universally called "12-6". Reading the two the other way round silently swaps the two coefficients, and since both are positive nothing downstream complains — the tuning still converges, onto the wrong labels.

case	potential	p_1 obtained	p_1 published	p_2 obtained	p_2 published	deviation
nn	well	1.109228	1.1096	0.391087	0.3918	0.18%
nn	Pöschl-Teller	0.907086	0.9071	0.799622	0.7991	0.07%
nn	Gaussian	1.211903	1.2121	0.567906	0.5672	0.12%
nn	Lennard-Jones	9.760657	9.8667	3.005560	3.0884	**2.68%**
unit.	well	1.233701	1.2337	1.000000	1.0000	0.00%
unit.	Pöschl-Teller	1.000000	1.0000	2.000000	2.0000	0.00%
unit.	Gaussian	1.342002	1.3420	1.435224	1.4349	0.02%
unit.	Lennard-Jones	0.264625	0.2646	0.000341	0.00034	0.00%
deut.	well	1.758644	1.7575	0.499215	0.5000	0.16%
deut.	Pöschl-Teller	1.423895	1.4388	0.886001	0.8631	**2.65%**
deut.	Gaussian	1.910904	1.9102	0.674817	0.6754	0.09%
deut.	Lennard-Jones	6.846862	6.8147	0.915045	0.9049	1.13%

The two rows in bold are the `D1` divergence: the published parameters achieve $r_0 = 2.71$ and 1.73, while Table 2 of the same paper asks for 2.70 and 1.70. Refining the grid does not move these deviations. The third row above 0.2%, (deut., Lennard-Jones) at 1.13%, is not of that kind: there the published parameters do reach the target, and the gap is the flat direction in (C_{12}, C_6) discussed above.